API Documentation

Base URL: http://127.0.0.1:8000. Routes below include the /api prefix. FastAPI publishes the interactive OpenAPI schema at /docs.

Request validation uses Pydantic. Invalid bodies or query constraints return HTTP 422 with FastAPI's detail list. Database and artifact availability affect runtime-dependent responses. JSON examples below show response shapes; dynamic values depend on the local data and artifacts.

GET /api/health

Checks database connectivity and reports artifact availability. No parameters or body. Returns HTTP 200 with status, database, chroma, bm25, langgraph, models, llm, and artifact_errors. status is ok or unavailable; the route itself returns 200 even when the database is unavailable.

Sample response shape:

json {"status":"ok","database":"available","chroma":"available","bm25":"available","langgraph":"available","models":{"production_impact":true},"llm":"LLM not configured","artifact_errors":{"retrieval":null,"workflow":null,"models":{}}}

GET /api/disruptions

Lists structured cases. Optional query parameters: disruption_type, industry, supplier_region, disruption_severity (1–5), limit (default 50, 1–500), and offset (default 0, minimum 0). No body. Returns HTTP 200 with total, limit, offset, and results. Database errors are not translated to custom responses.

Example: GET /api/disruptions?industry=Electronics&limit=2&offset=0

Response shape:
{"total":123,"limit":2,"offset":0,"results":[{"id":1,"disruption_id":"DIS-000001","disruption_type":"Factory Incident","industry":"Electronics","supplier_tier":"1","supplier_region":"Asia-Pacific","supplier_size":"Large","has_backup_supplier":true,"disruption_severity":4,"production_impact_pct":54.3,"revenue_loss_usd":11376013.16,"response_type":"Customer Delay","response_time_days":14,"partial_recovery_days":3,"full_recovery_days":59,"permanent_supplier_change":false,"created_at":"<database timestamp>"}]}. Result values are database-dependent.

GET /api/disruptions/{disruption_id}

Fetches a row by the database integer primary key (despite the parameter label, this is not the CSV string DIS-* identifier). Path parameter disruption_id: integer. Returns 200 and one disruption object, or 404 {"detail":"Disruption record not found"}. No request body.

Example: GET /api/disruptions/1.

POST /api/retrieval/similar

Searches the hybrid retriever. Body: query string (3–1200 characters), optional filters object, top_k (default 10, 1–50). Filters accept disruption type, industry, supplier tier, region, size, backup flag, severity (1–5 or $eq/$gte/$lte), and response type. Returns 200 with results, count, normalized filters, and configured weights. Empty matches return an empty results array. Returns 503 if retrieval artifacts are unavailable. Invalid input returns 422.

Sample request: {"query":"severe electronics supplier disruption in Asia-Pacific","top_k":5,"filters":{"has_backup_supplier":false}}.

Sample response shape: {"results":[{"disruption_id":"DIS-...","semantic_score":0.81,"keyword_score":0.64,"hybrid_score":0.76,"disruption_type":"Port Congestion","industry":"Electronics","supplier_tier":"2","supplier_region":"Asia-Pacific","supplier_size":"Medium","has_backup_supplier":false,"disruption_severity":4,"production_impact_pct":42.1,"revenue_loss_usd":1000000,"response_type":"Alternative Supplier","response_time_days":3,"partial_recovery_days":20,"full_recovery_days":61,"permanent_supplier_change":false}],"count":1,"filters":{"has_backup_supplier":false},"weights":{"semantic":0.7,"keyword":0.3}}. Scores/outcomes vary with actual indexes and weights.

POST /api/predictions

Runs persisted prediction models. Required body fields: disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity (1–5). Returns 200 with predictions, explanations, and features_used. Returns 503 Model not trained if model artifacts are unavailable; invalid requests return 422.

Sample request: {"disruption_type":"Port Congestion","industry":"Electronics","supplier_tier":"2","supplier_region":"Asia-Pacific","supplier_size":"Medium","has_backup_supplier":false,"disruption_severity":4}.

Response shape: {"predictions":{"production_impact":{"prediction":"<measured at runtime>","unit":"%"},"financial_loss":{"prediction":"<measured at runtime>","unit":"USD"},"recovery_time":{"prediction":"<measured at runtime>","unit":"days"},"permanent_supplier_change":{"prediction":"<measured at runtime>","probability":"<model score>"}},"explanations":{},"features_used":["disruption_type","industry","supplier_tier","supplier_region","supplier_size","has_backup_supplier","disruption_severity"]}.

POST /api/disruptions/analyze

Runs the full workflow and saves an analysis. Required fields: description (12–1200 chars), disruption type, industry, supplier tier, supplier region, supplier size, backup-supplier boolean, and severity 1–5. Optional observed production_impact_pct (0–100), revenue_loss_usd (0–1e12), response_type, and response_time_days (0–5000). Returns 200 with analysis_id, classification, priority, similar_cases, predictions, response_analysis, recovery_analysis, resilience_analysis, recommendation, evidence, explanation, validation, and llm_status. Returns 503 when workflow or retrieval is unavailable or execution fails. Invalid input returns 422.

Sample request:

json {"description":"A severe port disruption has delayed an electronics supplier in Asia-Pacific.","disruption_type":"Port Congestion","industry":"Electronics","supplier_tier":"2","supplier_region":"Asia-Pacific","supplier_size":"Medium","has_backup_supplier":false,"disruption_severity":4}

Response shape: {"analysis_id":1,"classification":{},"priority":{},"similar_cases":[],"predictions":{},"response_analysis":[],"recovery_analysis":{},"resilience_analysis":{},"recommendation":{"recommended_response":"<retrieved strategy or insufficient evidence>","reasoning":[],"confidence":0.0,"human_review_required":true},"evidence":[],"explanation":{},"validation":{"valid":false,"errors":[]},"llm_status":"LLM not configured"}. Dynamic fields depend on stored cases/models.

POST /api/recommendations

Returns a saved recommendation. Required query parameter analysis_id (>0). No body. Returns 200 with recommendation JSON, or 404 {"detail":"Recommendation not found"}. Invalid query returns 422.

Example: POST /api/recommendations?analysis_id=1.

GET /api/analytics/overview

Returns aggregate counts and averages from all ingested records. No parameters/body. Response 200 fields: total_disruptions, average_production_impact_pct, total_revenue_loss_usd, average_response_time_days, average_recovery_time_days, backup_supplier_coverage_pct, permanent_supplier_change_rate_pct.

GET /api/analytics/disruptions

Returns counts grouped by disruption type, industry, supplier region, and severity. No parameters/body. Response 200 shape: {"by_type":[{"label":"Port Congestion","count":25118}],"by_industry":[],"by_region":[],"by_severity":[]}. Counts depend on ingested rows.

GET /api/analytics/responses

Returns one row per observed response_type with case_count, median production impact, median revenue loss, median response time, median partial recovery, and median full recovery. No parameters/body. Returns HTTP 200 array; an empty database returns [].

GET /api/analytics/recovery

Returns full-recovery counts in fixed day buckets (0-14, 15-30, 31-60, 61-90, 91-180, 181+) and median_full_recovery_days. No parameters/body. Response 200 shape: {"distribution":[{"recovery_bucket":"0-14 days","case_count":123}],"median_full_recovery_days":61}.

POST /api/feedback

Stores feedback for an existing analysis. Required: positive integer analysis_id, boolean useful. Optional: actual_response, actual_recovery_days (0–5000), actual_production_impact_pct (0–100), actual_revenue_loss_usd (0–1e12), comment (up to 1000 chars). Returns 201 {"id":1,"message":"Feedback saved"}; 404 if the analysis does not exist; 422 for invalid input.

Sample request: {"analysis_id":1,"useful":true,"comment":"Evidence was relevant.","actual_recovery_days":42}.

GET /api/evaluation

Returns local model metrics from MODEL_METRICS_PATH when present and up to 50 stored DeepEval results. No parameters/body. Response 200 fields: model_metrics, evaluations, and status (available or Not available). A database failure can prevent the route from completing.

GET /api/models

Returns saved model availability and the local metrics report. No parameters/body. Response 200 fields: availability, metrics, and status (available or Model not trained).